

TreeGAN

Peter A C Nelson
peteracnelson@hkbu.edu.hk
Hong Kong Baptist University
Hong Kong



ABSTRACT

TreeGAN is an investigation into how machine learning and generative adversarial networks (GANs) create 3-dimensional objects. As machine learning finds an increasing number of applications within visual culture, we were interested to see how such systems might influence how we think about 3D objects. When this project started in 2019, there were relatively few art projects that used machine learning to produce 3D objects and even fewer that were trained on 3D objects to produce 3D objects (as opposed to synthesising 3D forms from 2D images), partly due to the paucity of conditional datasets of 3D objects. We synthesised a dataset of 3D objects using a form that is easy to produce and recognise - trees. Previous studies for 3D machine learning tended to focus on geometrically simple objects such as IKEA furniture (Lim et al 2013)

and industrial objects (Wu et al 2016), therefore, trees presented an opportunity to observe how a 3D machine learning system would approach complex yet familiar organic forms. Trees are often used in visual art as metaphors for the human experience, from the scholarly pines of Chinese ink painting (Clunas 2002) (McMahon 2003) to the martyred oaks of German Romanticism (Rosenblum 1975), and thus add an empathetic layer to our formal exploration. Three-dimensional trees are easy to produce on a large scale using Lindenmayer systems, and we made 76 unique tree templates, based on art historical references and exported 350 random variations of these templates, giving us a dataset of just over 26,000 3D trees. We watched the transition of beautiful abstractions as the system progressed from random 3D noise to recognizable trees, a process we likened to the analytical cubism of Picasso and Braque in the early 20th century, where we could observe a new technological system developing its own form of figuration.

ACM Reference Format:

Peter A C Nelson. 2023. TreeGAN. In *Proceedings of SA Art Gallery '23*. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3610537.3622953>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Art Gallery '23, December 12-15, 2023, Sydney, NSW, Australia

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0308-9/23/12.

<https://doi.org/10.1145/3610537.3622953>